

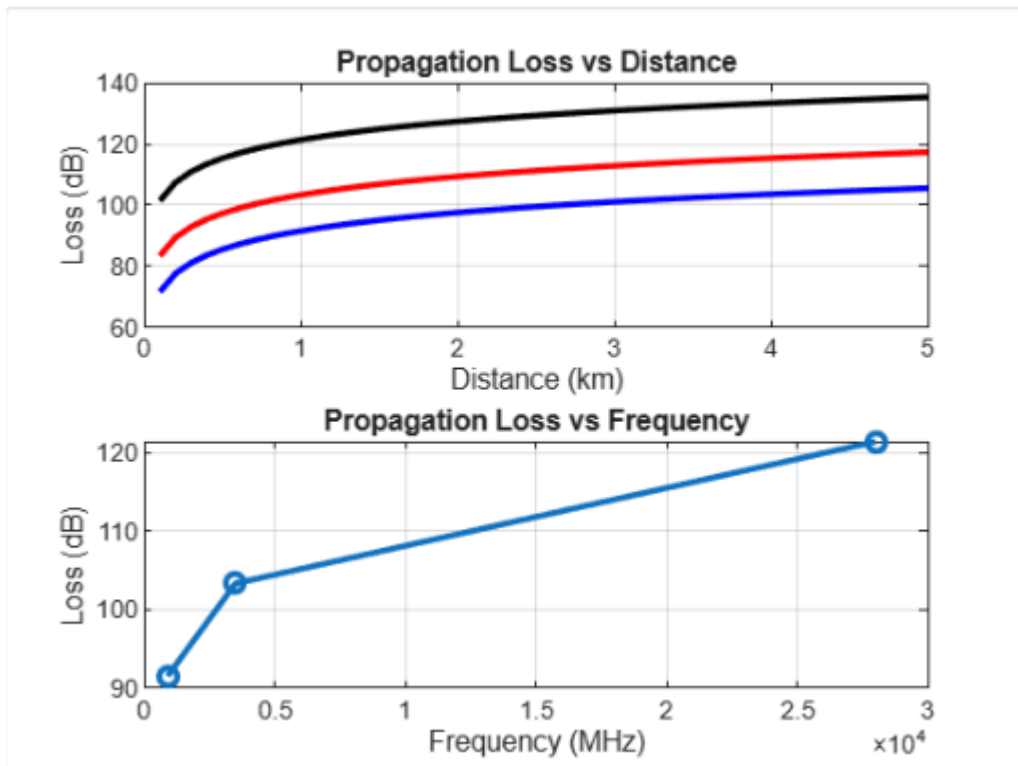
Ex.N.3: Spectrum Analysis of Sub-1 GHz, Mid-band & mmWave by Plotting Propagation

Loss. Date

Objective 1: Matlab Code and Output:

```
clc; clear; close all;

d = 0.1:0.1:5; % Distance (km)
f = [900 3500 28000]; % Frequency (MHz)
PL1 = 32.44 + 20*log10(f(1)) + 20*log10(d);
PL2 = 32.44 + 20*log10(f(2)) + 20*log10(d);
PL3 = 32.44 + 20*log10(f(3)) + 20*log10(d);
figure
% ----- Graph 1 -----
subplot(2,1,1)
plot(d,PL1,'b','LineWidth',2); hold on
plot(d,PL2,'r','LineWidth',2)
plot(d,PL3,'k','LineWidth',2)
title('Propagation Loss vs Distance')
xlabel('Distance (km)')
ylabel('Loss (dB)')
grid on
% ----- Graph 2 (NO legend used to avoid error) -----
subplot(2,1,2)
Freq = [900 3500 28000];
Loss = [PL1(10) PL2(10) PL3(10)];
plot(Freq,Loss,'-o','LineWidth',2)
title('Propagation Loss vs Frequency')
xlabel('Frequency (MHz)')
ylabel('Loss (dB)')
grid on
```



Objective2

`clc; clear; close all;`

`d = 2; % fixed distance (km)`

`f = [900 3500 28000]; % MHz`

`PL = 32.44 + 20*log10(f) + 20*log10(d);`

`figure`

`% ----- Graph 1: Frequency vs Loss -----`

`subplot(2,1,1)`

`plot(f,PL,'-o','LineWidth',2)`

`title('Propagation Loss vs Operating Frequency')`

`xlabel('Frequency (MHz)')`

`ylabel('Propagation Loss (dB)')`

`grid on`

`% ----- Graph 2: Band Comparison -----`

`subplot(2,1,2)`

`bar(PL)`

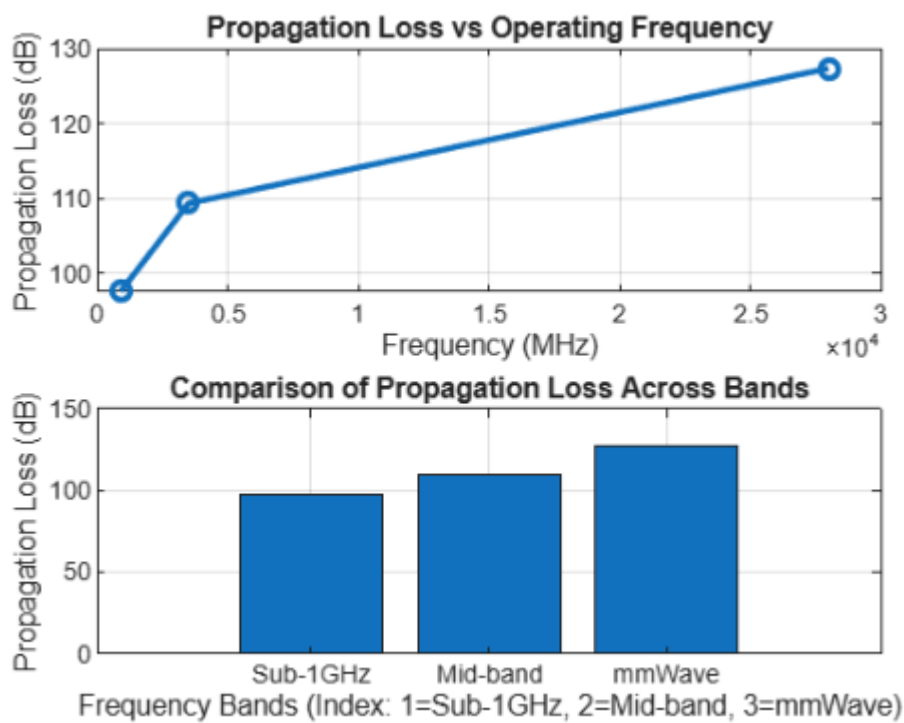
```
title('Comparison of Propagation Loss Across Bands')
```

```
xlabel('Frequency Bands (Index: 1=Sub-1GHz, 2=Mid-band, 3=mmWave)')
```

```
ylabel('Propagation Loss (dB)')
```

```
xticklabels({'Sub-1GHz', 'Mid-band', 'mmWave'})
```

```
grid on
```



Objective 3

```
clc; clear; close all;
```

```
% Frequency bands (MHz)
```

```
f = [900 3500 28000];
```

```
% Distance (km)
```

```
d = 2;
```

```
% Propagation Loss (FSPL)
```

```
PL = 32.44 + 20*log10(f) + 20*log10(d);
```

```
figure
```

% ----- Graph 1: Simple Line Plot (SAFE) -----

```
subplot(2,1,1)
```

```
plot(f, PL, '-o', 'LineWidth', 2)
```

```
title('Propagation Loss vs Operating Frequency')
```

```
xlabel('Frequency (MHz)')
```

```
ylabel('Propagation Loss (dB)')
```

```
grid on
```

% ----- Graph 2: Safe Comparison Plot -----

```
subplot(2,1,2)
```

```
plot(1:3, PL, '-s', 'LineWidth', 2)
```

```
title('Band-wise Propagation Loss Comparison')
```

```
xlabel('Band Index (1=Sub1GHz, 2=Mid-band, 3=mmWave)')
```

```
ylabel('Propagation Loss (dB)')
```

```
grid on
```

